AI Methods for Turbulence and Early-Universe Signals

A KAAI Workshop · Carnegie Mellon University

November 2 – 6 · McWilliams Center for Cosmology & Astrophysics, CMU



SIMONS FOUNDATION

McWilliams Center for Cosmology & Astrophysics

OVERVIEW

About the workshop

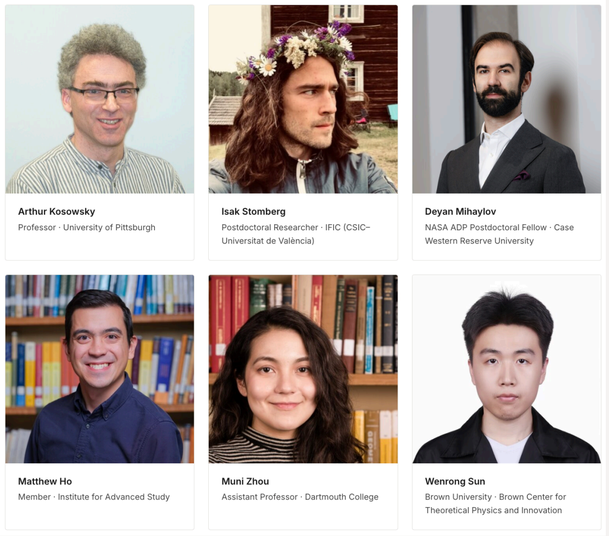
Turbulence appears across astrophysics and cosmology, from magnetized plasmas in galaxies and the interstellar medium to the primordial plasma of the early universe. Early-universe processes such as first-order phase transitions and primordial magnetic fields can drive turbulence, which may leave observable imprints in stochastic gravitational-wave backgrounds and in cosmic magnetic fields. Modeling these systems requires expensive simulations over a wide range of scales, and connecting them to data is a hard inverse problem. This workshop brings together researchers in turbulence, early-universe physics, and machine learning to discuss how AI methods can help with simulation, inference, and signal detection in this area.

- **AI methods for turbulence:** surrogate models, emulators, and learned representations of hydrodynamic and magnetohydrodynamic turbulence across scales.
- **Early-universe signals:** modeling and inference of signatures of early-universe physics, including gravitational-wave backgrounds and primordial magnetic fields.

This is a Keystone Astronomy & AI (KAAI) Fellows Program workshop, sponsored by the Simons Foundation Targeted Grant to Institutes.

- Talks on: early Universe physics, GW production and detection, MHD turbulence simulations, plasma physic simulations, AI for cosmology, machine learning
- Crash course on Pencil Code for MHD simulations and GW production
- Time for open discussion / collaborative projects
- Hackathon and open-problem challenges

Invited speakers



09:30	Welcome and Introductions	Wenrong Su	Touko Puro / Shreyas Giridhar	Chris Choi	Hackathon
09:45					
10:00	Alberto Roper Pol		Isak Stomberg*	James Wedgwood	
10:15		Crash course on numerical simulations with Pencil Code			
10:30					
10:45	Coffee break	Coffee break	Coffee break	Coffee break	Coffee break
11:00					
11:15	Arthur Kosowsky	Crash course on numerical simulations with Pencil Code	Deyan Mihaylov	Muni Zhou	Hackathon
11:30					
11:45					
12:00	Lunch	Lunch	Lunch	Lunch	Lunch
12:15					
12:30					
12:45					
13:00	Tina Kahniashvili	Crash course on numerical simulations with Pencil Code	Matthew Ho	Free time to discuss / work on collaborative projects	Hackathon
13:15					
13:30					
13:45	Introduction of the proposed projects for turbulence	Project proposal with Pencil Code	Free time to discuss / work on collaborative projects		
14:00					
14:15					
14:30	Open discussion about projects				
14:45					
15:00					
15:15	Murman Gurgenidze				
15:30					
15:45					
16:00	Coffee break	Coffee break	Coffee break	Coffee break	Coffee break
16:15					
16:30	Free time to discuss / work on collaborative projects	Ethical concerns about use of AI	Free time to discuss / work on collaborative projects	Open discussion about future of science with AI	Closing / winners of hackathon
16:45					
17:00					
17:15	Free time to discuss / work on collaborative projects				
17:30					
17:45					

The Pencil Code

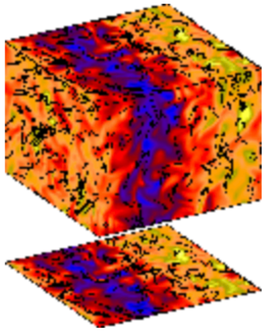
a modular multi-purpose Fortran+MPI code for differential equations and particles

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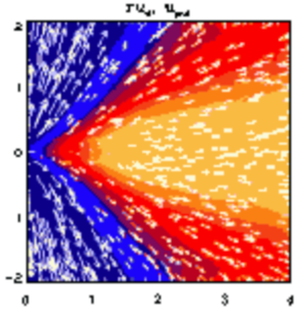
SCAN ME

The **Pencil Code** a modular MPI code for differential equations and particles: multipurpose and multiuser-maintained. It is highly modular and can easily be adapted to different types of problems. The code runs efficiently under MPI on massively parallel shared- or distributed-memory computers.

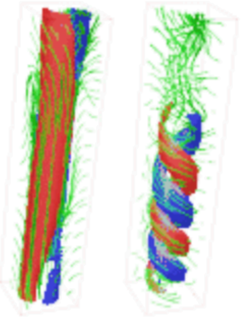
The Pencil Code or equivalent codes have been used for many different applications in a (more or less) astrophysical context. Examples are



Turbulence simulations



Outflows from accretion discs



Dynamo experiments

Available as open source code on **https://github.com/pencil-code/pencil-code/**. It is being updated after each change in the developer's repository, which can be accessed both through svn and git; see **https://pencil-code.org/download.php**.

Pencil News

Recent news item about the Newsletter, Pencil Code office hours, and the the Pencil Code Steering Committee: [more...]

Get Pencil

There are several ways how to get the code. [more...]

Learn Pencil

Quick start guide fo beginners, samples, manual & [more...]

PC school material

Accelerating the Pencil Code

CERN

Pencil Code User Meeting

October 2025 - CERN (Switzerland)

1st Pencil Code school on early Universe physics and gravitational waves

Lectures and course material here



Scan me to submit questions during the workshop!

Tuesday discussion: Ethical concerns about use of AI

Thursday discussion: Future of science with AI